

Control of meristem determinacy by trehalose 6-phosphate phosphatases is uncoupled from enzymatic activity

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Meristem fate is regulated by trehalose 6-phosphate phosphatases (TPPs), but their mechanism of action remains mysterious. Loss of the maize TPPs *RAMOSA3* and *TPP4* leads to reduced meristem determinacy and more inflorescence branching. However, analysis of an allelic series revealed no correlation between enzymatic activity and branching, and a catalytically inactive version of *RA3* complements the *ra3* mutant. Together with their nuclear localization, these findings suggest a moonlighting function for TPPs.

The body plan of plants is determined by meristems, self-renewing pools of pluripotent stem cells that produce new structures such as roots, leaves and floral organs. Meristem development is tightly controlled by multiple pathways¹, and an important regulator of meristem fate is the phosphorylated disaccharide trehalose 6-phosphate (T6P). For example, loss-of-function mutations in *TREHALOSE 6-PHOSPHATE SYNTHASE 1* (*TPS1*), the major enzyme for T6P biosynthesis in *Arabidopsis* (*Arabidopsis thaliana*), are embryo lethal, and when rescued during embryogenesis, impair the transition to reproductive development². In maize (*Zea mays*), *RAMOSA3* (*RA3*) encodes a trehalose 6-phosphate phosphatase (TPP) that dephosphorylates T6P to trehalose³. *ra3* mutants have reduced meristem determinacy, leading to increased tassel branching and ectopic branching in ears³. Despite its central role in controlling growth and development, knowledge on how T6P functions is limited. T6P levels correlate with sucrose levels and are thought to be a signal that maintains appropriate sucrose levels for the tissue and developmental stage⁴. However, the molecular targets of T6P remain largely elusive. Sucrose non-fermenting 1-related kinase 1 functions in T6P-mediated growth control⁵, but the extent to which T6P signals through additional pathways is a subject of debate^{4,6}. A better mechanistic understanding of T6P could increase crop yields, as expression of a rice TPP in developing maize ears enhances yields under drought⁷, and treatment with plant-permeable analogues of T6P increases grain yield and drought tolerance in wheat⁸.

To better understand how TPPs regulate meristem determinacy, we isolated genetic modifiers by mutagenizing *ra3* mutants using ethyl methanesulfonate (EMS) and screening for enhanced ear branching. In this screen, we identified four independent lines with mutations in the *RA3* paralogue *TREHALOSE 6-PHOSPHATE PHOSPHATASE 4* (*TPP4*, Zm00001d052227) that were associated

with increased ear branching (Fig. 1a,b). All four alleles had missense mutations in residues that were conserved among plant TPPs (Fig. 1c and Supplementary Fig. 1). We confirmed that *TPP4* was the causal gene by generating a null allele using clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated protein 9 (Cas9), which enhanced the *ra3* phenotype to the same extent as our EMS alleles (Supplementary Fig. 2). *tpp4* mutants had normal, unbranched ears in the presence of functional *RA3*, but in a *ra3*-mutant background, *tpp4* semidominantly increased ear and tassel branch number (Fig. 1a,b and Supplementary Fig. 3) and branches extended up the ear rather than being restricted to the base as in *ra3* single mutants (Fig. 1a). Thus, *TPP4* is a fully redundant paralogue of *RA3* and, accordingly, its expression was lower than *RA3* in developing ears⁹. However, *TPP4* was strongly upregulated in *ra3* mutants (Fig. 1d), suggesting that it compensates for the loss of *RA3* in a responsive backup circuit¹⁰. In situ hybridization with *TPP4*-specific probes showed that this compensation occurred specifically in the expression domain of *RA3*, which subtends spikelet pair meristems³ (Fig. 1e). An additional TPP, *TPP12*, was also upregulated in developing *ra3*-mutant ears (Fig. 1d and Supplementary Fig. 4), but CRISPR–Cas9-induced *tpp12* mutants did not enhance *ra3* ear branching (Supplementary Fig. 4), indicating that *RA3* and *TPP4* are the major TPPs affecting meristem determinacy.

To further investigate how TPPs affect inflorescence development, we measured T6P levels in developing ears of *ra3* and *ra3;tpp4-1* double mutants at the stage during which *RA3* and *TPP4* expression peaks. Surprisingly, there was no difference in T6P levels compared to the B73 wild type (Fig. 2a). Although it is possible that localized effects of loss of *RA3* and *TPP4* were masked by the analysis of entire ear primordia, we speculated that *TPP4* enzymatic activity may not be important, consistent with the fact that our EMS-generated *tpp4* alleles did not affect motifs important for phosphatase activity. We modelled *TPP4* structure using structures of similar TPPs, and only the Ser167 residue mutated in *tpp4-2* was close to the active site (Fig. 2b). We also predicted that mutating Ala325 to Thr in *tpp4-1* would result in steric clash with the nearby highly conserved Phe302 residue and disrupt the catalytic pocket. However, residues mutated in *tpp4-3* and *tpp4-4* were far from the active site on exposed surface loops, where their effect on protein structure and activity was not obvious (Fig. 2b). To test our structural predictions, we asked whether

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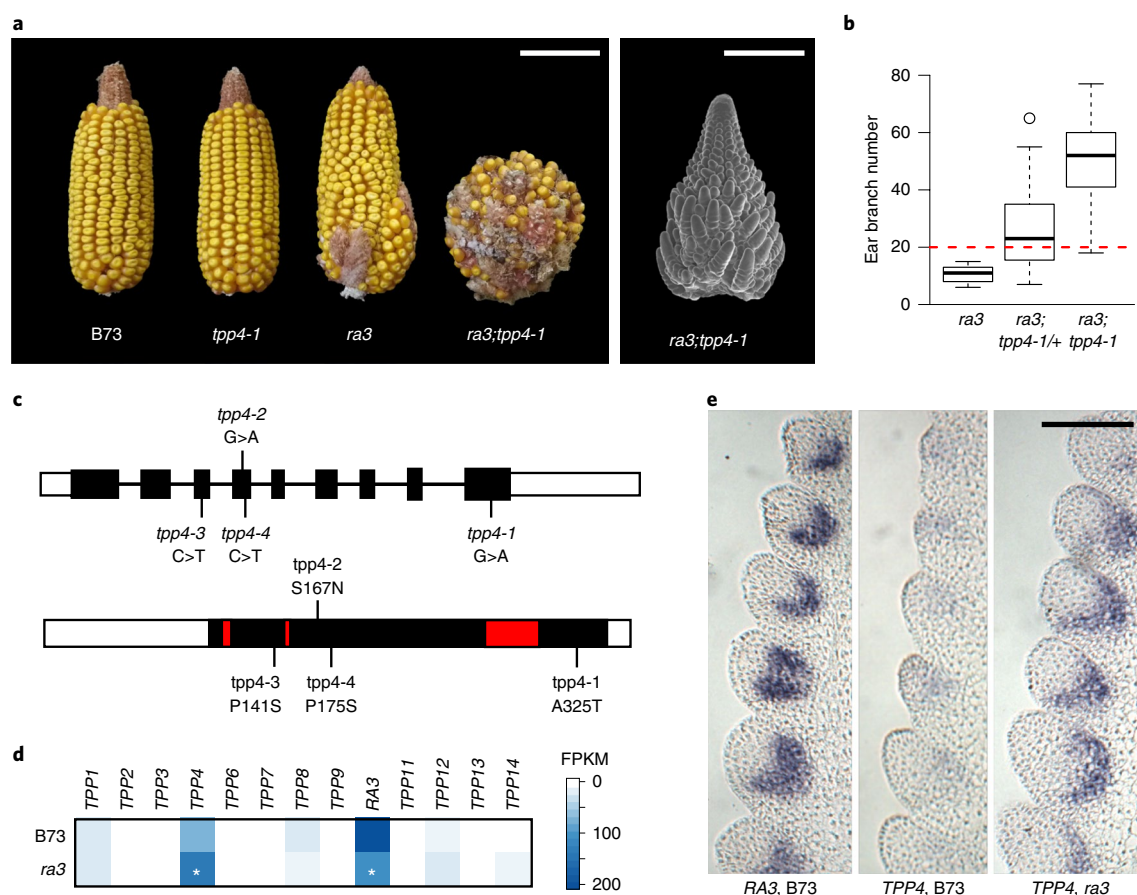


Fig. 1 | TPP4 acts as a redundant backup for RA3. **a**, Representative mature ears of B73 wild type (WT), *tpp4-1*, *ra3* and *ra3;tpp4-1*, with enhanced branching compared to *ra3*. Phenotypes were consistent across all (≥ 5) growing seasons. Scale bar, 10 cm. A representative electron micrograph of a developing *ra3;tpp4-1* ear on the right shows branching extending up the ear. Ears from ten plants were imaged, with similar results. Scale bar, 1 mm. **b**, Co-segregation of *tpp4-1* with ear branch number in a *ra3*-mutant background. The dashed red line indicates our arbitrary threshold for enhancement (20 ear branches). $n = 11, 40$ and 13 , respectively. The bottom and top of the boxes are the first and third quartiles, respectively, the middle line is the median, the whiskers are the most extreme values within $1.5 \times$ the interquartile distance below or above the first or third quartile, and the remaining outlier data points are shown as individual dots. **c**, Overview of EMS-induced *tpp4* alleles. Top, the position of mutations in *TPP4*, with boxes indicating exons (the coding sequence is in black) and the lines indicating introns. Bottom, the positions of amino acid substitutions in the TPP4 protein, with the solid black portion indicating the TPP domain and conserved phosphatase boxes in red. **d**, Heatmap of expression levels of all maize TPPs in developing ears in WT B73 and *ra3* mutants. *TPP5* is a pseudogene and was excluded. The asterisks indicate differential expression between B73 and *ra3* ears ($P < 0.05$, exact negative binomial test in edgeR with Benjamini–Hochberg multiple testing correction; $n = 3$ biological replicates). *TPP4* was significantly upregulated in *ra3* (2.4-fold, $P = 0.040$) and *RA3* was significantly downregulated (2-fold, $P = 0.012$). FPKM, fragments per kilobase of transcript per million mapped reads. **e**, In situ hybridization using an antisense *RA3* probe in a WT ear, showing expression subtending developing spikelet pair meristems, using an antisense *TPP4* probe in a WT ear or using a *ra3*-mutant ear. *TPP4* is expressed in the same domain as *RA3* and is stronger in *ra3* mutants. Representative images are shown; the experiment was performed twice with similar results, using ears from at least three independent plants per genotype per experiment. Scale bar, $100 \mu\text{m}$.

the proteins encoded by the different *tpp4* alleles could complement the *Saccharomyces cerevisiae* *tpp* mutant $\Delta tps2$ (ref. ¹¹). Expression of wild-type *TPP4* rescued the $\Delta tps2$ growth defect, whereas a negative control, *tpp4*^{NYN}, in which two catalytically important Asp residues were mutated to Asn, did not, suggesting that enzymatic activity was required to complement the yeast mutant (Fig. 2c). We next tested our *tpp4* alleles, and although *tpp4-1* was unable to complement $\Delta tps2$, partial complementation was observed for *tpp4-2*, *tpp4-3* and *tpp4-4*, suggesting that they maintain some level of activity, confirming our structural predictions (Fig. 2c). To validate these results and to measure enzymatic activity, we used purified fusion proteins with maltose-binding protein for in vitro assays. Similar to *RA3* (ref. ³), *TPP4* activity was specific for T6P and did not dephosphorylate other sugar phosphates (Supplementary Fig. 5). We quantified the

degree of activity for different *tpp4* alleles and found that, although *tpp4-1* had no detectable activity, *tpp4-2* had low activity, and *tpp4-3* and *tpp4-4* had significant activity at $\sim 25\%$ and $\sim 35\%$ of the wild-type levels, respectively (Fig. 2d).

We next used our *tpp4* allelic series to ask whether the branching phenotype correlated with enzymatic activity. All alleles were backcrossed to *ra3* in B73 at least three times, and we phenotyped segregating populations grown in replicate in three different field locations. The degree of ear branching was normalized across locations using segregating controls to account for the influence of the environment on the phenotype. All *tpp4* alleles enhanced *ra3* semidominantly; however, despite the differences in enzyme activity, all alleles affected ear branching to the same degree, resulting in a twofold increase in *ra3;tpp4* heterozygotes (Spearman's

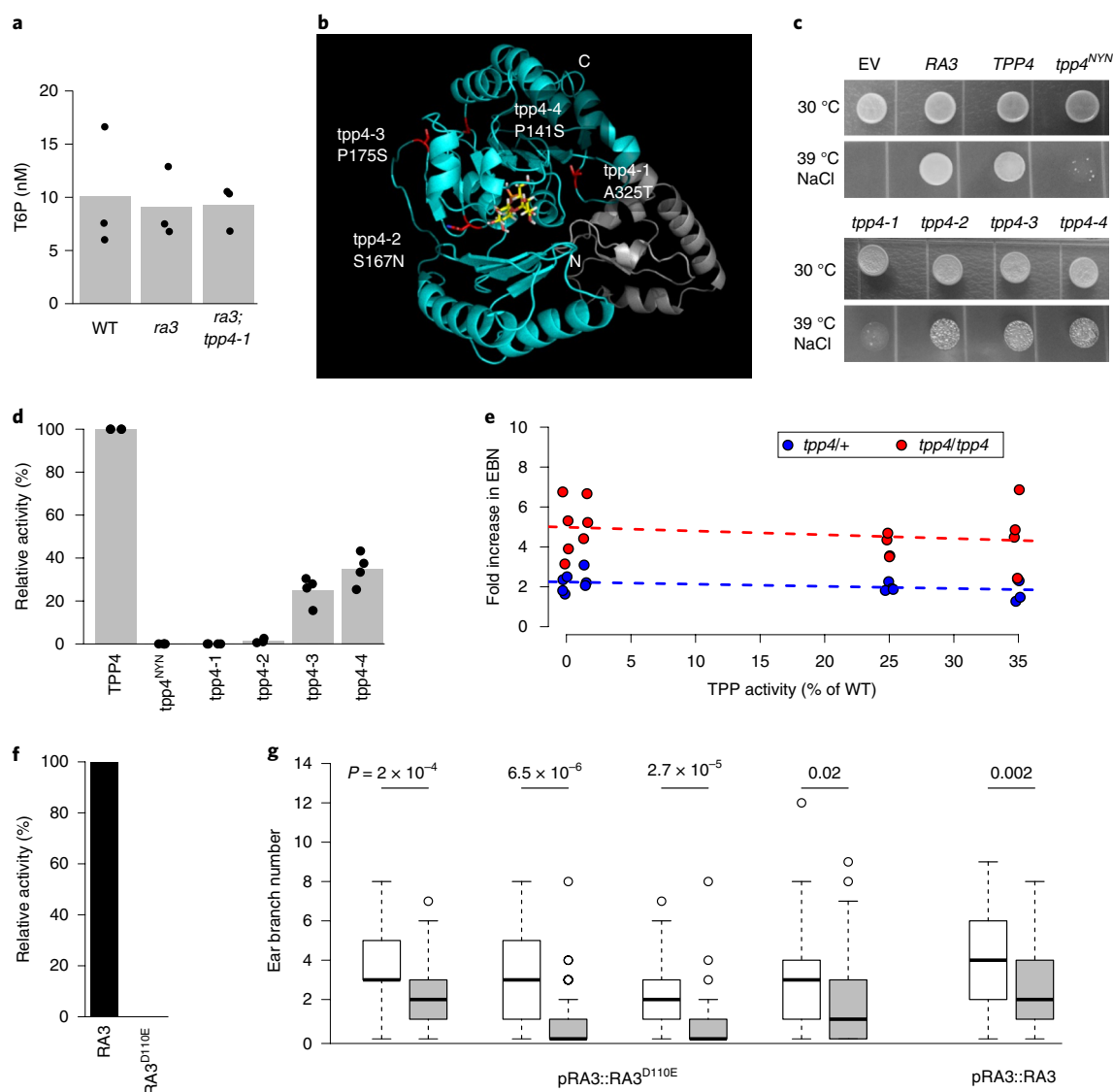


Fig. 2 | TPP enzymatic activity does not correlate with ear branching phenotype. **a**, The levels of T6P are unchanged in *ra3* or *ra3;tpp4-1* mutants ($P=0.80$ for the B73-*ra3* comparison; $P=0.82$ for the B73-*ra3;tpp4-1* comparison; two-sided Student's *t*-test). Data points from three biological replicates are shown as dots; the bars indicate the mean. **b**, Homology model of TPP4: the N-terminal extension is shown in grey and the TPP domain is in cyan, the N and C termini are labelled and T6P is modelled in the active site in yellow. Residues changed in the EMS mutants are shown in red, with all but Ser 167 (*tpp4-2*) located on the protein surface. T6P is modelled in the active site in yellow. **c**, RA3, TPP4 and most *tpp4* EMS alleles complemented the yeast *tpp* mutant $\Delta tps2$, whereas catalytically inactive *tpp4^{N/N}* did not. EV, empty vector. Images from a representative experiment are shown; two independent experiments were performed with similar results. **d**, *tpp4-3*-mutant and *tpp4-4*-mutant proteins have considerable TPP activity. The dots indicate data points from four independent biological replicates; the bars show the mean. **e**, Relative increase in ear branch number in *ra3;tpp4* heterozygous (blue) and *ra3;tpp4* homozygous (red) mutants compared to segregating single *ra3* mutants, plotted against TPP activity of the different alleles (from left to right: *tpp4-1*, *tpp4-2*, *tpp4-3* and *tpp4-4*), showing no correlation between enzymatic activity and ear branching (Spearman's correlation = -0.20 , $P=0.47$ for *tpp4* heterozygotes; Spearman's correlation = -0.08 , $P=0.76$ for homozygotes; *P* values were calculated using a two-sided Student's *t*-test with asymptotic *t* approximation). The dots indicate data points from four independent biological replicates, with linear trend lines shown as dashed lines. **f**, RA3^{D110E}-mutant protein had no detectable TPP activity in vitro. **g**, *ra3* mutants were complemented by catalytically inactive RA3^{D110E}. Data are from segregating populations with siblings without the transgene in white and with the transgene in grey. The degree of complementation is similar for RA3^{D110E} and WT RA3 constructs. Box plots are defined as in Fig. 1b. The *P* values for each comparison were calculated using a two-sided Student's *t*-test; $n=45$, 55, 39, 53, 51, 44, 43, 43, 46 and 49.

correlation = -0.20 , $P=0.47$) and a fivefold increase in *ra3;tpp4* homozygotes (Spearman's correlation = -0.08 , $P=0.76$; Fig. 2e and Supplementary Fig. 6). These results strongly suggested that TPP enzymatic activity is not important for the control of branching.

To further test this hypothesis, we engineered a version of RA3 with a single amino acid change (RA3^{D110E}), which was predicted to maintain overall protein structure while abolishing catalytic

activity (Fig. 2f). Expression of this catalytically dead RA3 using its native promoter complemented the *ra3* phenotype, leading to a ~50% reduction in ear branch number compared to control siblings without the transgene (Fig. 2g). Similar partial complementation was seen in families expressing a wild-type version of RA3 from the same promoter (Fig. 2g), suggesting that the partial complementation was due to insufficient promoter elements in our constructs.

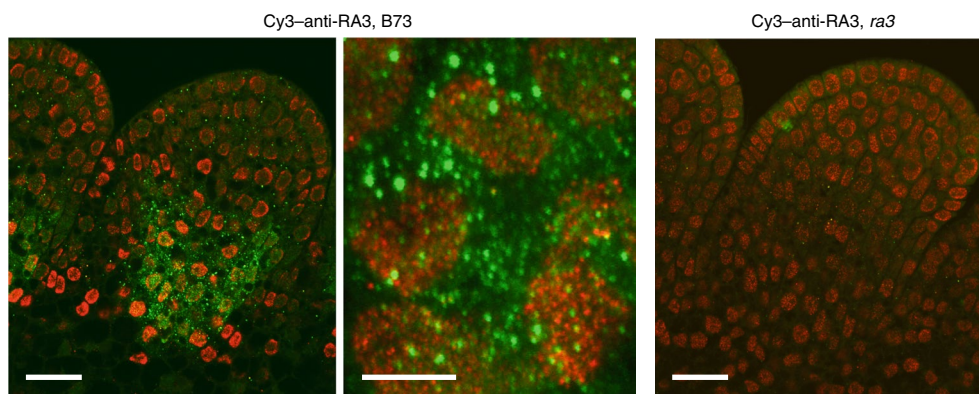


Fig. 3 | RA3 localizes to nuclear and cytoplasmic speckles. RA3 was detected by immunolocalization in nuclear and cytoplasmic speckles at the base of a spikelet pair meristem. RA3 is in green and nuclei are counterstained in red. The absence of a signal in the *ra3* mutant confirmed the specificity of the antibody. Representative images are shown; similar results were obtained in all five independent experiments. Scale bars, 25 μ m (left and right panels) and 5 μ m (middle panel).

Complementation of *ra3* by a catalytically dead RA3 suggests that a non-enzymatic function is responsible for its role in meristem determinacy.

Regulatory functions were postulated for the TPP family in *Arabidopsis*, as its expansion is due to whole-genome duplications with a high degree of paralogue retention, typical of regulatory proteins with stricter gene dosage requirements than enzymes¹². A regulatory function was also proposed for rice TPP7, based on its low activity and lack of effect on T6P levels¹³. Some metabolic enzymes are known to have regulatory functions, a phenomenon known as moonlighting¹⁴, and two other sugar metabolic enzymes, HEXOKINASE 1 and the fructose 1,6-bisphosphatase FRUCTOSE INSENSITIVE 1, exhibit moonlighting activity in *Arabidopsis*^{15,16}. As HEXOKINASE 1 regulates transcription in the nucleus¹⁵, we investigated the subcellular localization of RA3 and TPP4. Both proteins were predominantly localized in the nucleus when transiently expressed in tobacco (Supplementary Fig. 7) and we confirmed this by immunolocalization using RA3-specific antibodies in developing maize ears (Fig. 3). Interestingly, RA3 localized to speckles in cytoplasmic and nuclear compartments (Fig. 3) and these are typically associated with regulation of RNA processing and transcription¹⁷. As TPPs lack DNA-binding or RNA-binding domains, it is likely that their hypothetical nuclear role requires interaction with other proteins, consistent with our observation that mutations on the surface of TPP4 impair its role in meristem determinacy. Although the exact mechanism remains to be resolved, exploiting the dual roles of TPPs may open avenues for further crop improvement. TPP4 itself maps to a quantitative trait locus and near genome-wide association single-nucleotide polymorphisms (SNPs) for tassel branch number¹⁸ and is in a region that shows strong evidence of selection during maize domestication and improvement¹⁹, making it an especially attractive target for improvement of maize productivity.

Methods

Mapping. Pollen mutagenesis was performed as described²⁰ on *ra3-fea1* in B73 and *ra3-NI* in Mo17, and ~1,500 M2 families were screened for enhanced branching. To generate F2 mapping populations, enhancer candidates were crossed to *ra3-fea1* in the W23 or B73 background, and the resulting F1s were selfed. For each mutant, two pools were created using DNA from about 30 F2 individuals with either enhanced or non-enhanced ear branching, and SNP genotyping was performed using MaizeSNP50 BeadChip microarrays (Illumina). After removal of low-quality SNPs (GC > 0.6) and non-polymorphic SNPs (allele frequency < 0.2 or allele frequency > 0.8 in both pools), smoothed allele frequencies and their ratios were calculated using R 3.1.0 to find peaks associated with enhanced branching. Pooled genomic DNA from enhanced M2 individuals was used to make a library for next-generation sequencing using the NEBNext Ultra DNA Library Prep Kit for Illumina (New England Biolabs), and the sample was sequenced on a single

lane of Illumina HiSeq2500 to 10 $\times$ coverage. Reads were mapped to the maize B73 reference genome (AGPv3, <https://www.maizedb.org/assembly>) using bwa-mem²¹, and SNPs were called using the GATK unified genotyper²². High-confidence SNPs supported by at least three reads were selected if they were homozygous and prioritized if they were G>A or C>T transitions and in exons within the mapping interval. To confirm the mutation and detect causal mutations in other alleles, TPP4 was amplified from genomic DNA of M2 populations of additional enhancer lines for Sanger sequencing.

Generation of CRISPR–Cas9-induced mutants. Guide RNAs (gRNAs) targeting *TPP4* and *TPP12* were designed using CRISPR-P²³ (<http://cbi.hzau.edu.cn/crispr/>), and an array containing four gRNAs expressed from independent maize U6-derived promoters²⁴ was synthesized, with gRNAs GCAGATTGCCAGCGCGTCCCC and GTCGAGGTACCGCAGGGACA targeting *TPP4* and gRNAs GTAGTGTCAAGCAAAGCCC and GTGATCTTGCTGGGAGTAAA targeting *TPP12*. This array was cloned into the pMCG1005-Cas9 binary vector and transformed into the Hi-II background²⁵. Mutations were identified by Sanger sequencing and mutants were backcrossed at least three times to *ra3-fea1* (B73) before phenotyping. The inserted T-DNA containing Cas9 and the gRNA array was eliminated after the first backcross to ensure the stability of mutations.

Generation of transgenic complementation lines. To generate pRA3::RA3^{D110E} lines, the GAC codon encoding D110 was changed to GAG using site-directed mutagenesis of a RA3 genomic construct that spanned from 3,311 bp upstream of the ATG to the end of the 3' untranslated region. To generate control pRA3::RA3-FLAG-HA lines, a sequence encoding a FLAG tag followed by a triple haemagglutinin (HA) tag was introduced just before the stop codon of RA3 in the same genomic construct. The resulting constructs were cloned into the binary vector pAM1006 and transformed into Hi-II. Transgenic lines were backcrossed at least five times to *ra3-fea1* (B73) before phenotyping. Individuals carrying transgenes were identified by scoring resistance to the selectable marker and/or by PCR.

Phenotyping. For ear and tassel phenotyping, plants were grown in field locations in Cold Spring Harbor, NY, USA, or Lloyd Harbor, NY, USA (June–September), or Valle De Banderas, Nayarit, Mexico (November–February) between 2014–2018. Ears and tassels were collected after anthesis to count the branches. For scanning electron microscopy, developing ears were dissected around 6 weeks after planting and mounted on stubs to image on a Hitachi S-3500N scanning electron microscope. All phenotyping experiments were performed on segregating populations and each individual was genotyped.

Gene expression analysis. Ears (2–3 mm) were dissected from 6-week-old plants from a population segregating for *ra3* and *tpd4-1* and immediately frozen in liquid nitrogen. Tissue was ground using tungsten beads in a mixer mill and total RNA was extracted using the DirectZol RNA Miniprep Kit (Zymo Research), according to the manufacturer's instructions, including on-column digestion of genomic DNA with DNase I. Complementary DNA synthesis was performed using the iScript cDNA Synthesis Kit (Bio-Rad). Quantitative real-time PCR was performed using Universal SybrGreen Master Mix (Bio-Rad) on the CFX96 Real-Time System (Bio-Rad). Primers GGTCCGTCTCTGTTATTGATTG and ATTCCATCACCTCAGCTGGA were used for quantitative real-time PCR detection of *TPP12*. *UBIQUITIN 1* (*UBQ1*, Zm00001d015327) was

used for normalization (primers CCGCTTCAAGATGCAGATCTTTG and GAGACGGAGCACAAGGTGG).

RNA sequencing. Raw reads from previously described samples⁹ were trimmed and filtered using Trimmomatic 0.36, and paired reads were aligned against maize AGPv4.39 using HiSat2 2.1.0 (ref. 26). Counts were generated using HTSeq 0.6.0 (ref. 27) for all transcripts in the AGPv4.39 annotation, and differential expression was assessed using edgeR 3.22.0 (ref. 28) in R 3.5.0 after filtering for genes with c.p.m. > 1 in *N* samples, in which *N* represents the number of biological replicates. Members of the TPP family were identified using Gramene (<http://www.gramene.org>), and the two members that were not previously described²⁹ were named *TPP13* (Zm00001d006375) and *TPP14* (Zm00001d029371).

In situ hybridization. Developing ears (2–3-mm stage) were dissected from 6-week-old field-grown plants, followed by fixing, embedding and in situ hybridization as described³⁰, with 16 h of substrate incubation. *RA3* probes were as previously reported³, and for *TPP4*, two fragments were amplified from cDNA using primer pairs GCCTACATGAGCGACGTGAT/CCTCTCTCAGCACCTTGA and GTTGGGACGATCGAGAAAGT/GGCGTAGTAGAGCTCCGACA and subcloned into pCR2.1. Clones carrying inserts in both orientations were identified and sequence verified to generate antisense or sense probes by in vitro transcription with T7 polymerase (Roche).

In vitro TPP activity assays. *RA3*, *TPP4* and *TPP12* coding sequences were cloned into pMAL-c5x (New England Biolabs) and site-directed mutagenesis was performed on pMAL-c5x-*TPP4* to introduce the various point mutations. All sequences were verified by Sanger sequencing before transformation into the Rosetta *Escherichia coli* strain. Cultures were grown to an optical density at 600 nm (OD_{600}) of 0.6 at 37 °C, cooled to 16 °C before the addition of isopropyl- β -D-1-thiogalactopyranoside to a final concentration of 0.5 mM and grown for an additional 12–16 h at 16 °C. Culture harvesting and purification of MBP-TPPs were performed with the pMAL expression system (New England Biolabs) according to the manufacturer's instructions. Protein purity and concentration were assessed using SDS-PAGE gels. For in vitro assays, equal amounts of purified TPPs were incubated for 30 min at 28 °C in buffer containing 10 mM Tris-HCl, pH 7.6, 0.2 mM EDTA, 5 mM MgCl₂, 0.1 mg ml⁻¹ BSA and 0.5 mM T6P, sucrose 6-phosphate, glucose 6-phosphate or fructose 6-phosphate (Sigma-Aldrich). Phosphate release was measured using a colorimetric assay at OD_{400} using the Serine/Threonine Phosphatase Assay System (Promega). Activity was expressed as a percentage of the activity measured with purified recombinant *TPP4*.

Metabolite measurements. Developing ears (2–3 mm) were dissected from 7-week-old field-grown plants and immediately frozen in liquid nitrogen. T6P was extracted with chloroform-methanol and measured by high-performance anion-exchange chromatography coupled to tandem mass spectrometry⁶.

Protein structure homology model. The Phyre2 web portal for protein modelling and prediction³¹ was used to construct a homology model from the amino acid sequence of *TPP4*. The intensive search function of the web portal selected model templates 5GVX, 5DX9, 5DXE, 5HVO, 5LQD and 3T5T, corresponding to *Mycobacterium tuberculosis* TPP³², the *Cryptococcus neoformans* TPP (D24N)-T6P complex³³, the *Candida albicans* TPP amino-terminal domain³³, the *Aspergillus fumigatus* TPS in complex with UDP and valdioxylamine A³⁴, the *Streptomyces venezuelae* TPS³⁵ and a structurally similar transferase from *Streptomyces hygroscopicus* (unpublished), respectively. The TPP structures were necessary for modelling the phosphatase domain of *TPP4*, whereas the other structures, including the TPS structures, were incorporated for modelling the N-terminal domain of *TPP4*. From these templates, 96% of the residues were modelled at >90% confidence.

Yeast complementation. *RA3* and *TPP4* coding sequences were cloned into pYX212 (ref. 12) and site-directed mutagenesis was performed to introduce the various point mutations. All sequences were verified by Sanger sequencing before transformation into the *S. cerevisiae* YSH448 Δ tps2 strain. Clones were grown in liquid culture in SD-Ura medium to an OD_{600} of 1, spotted on SD-Ura plates with and without 1 M NaCl, and grown at either 30 °C or 39 °C for 2 d. Expression of TPP-HA fusion proteins was assessed by western blot using monoclonal anti-HA antibody from mouse (clone HA-7, Sigma-Aldrich).

Tobacco infiltration. *RA3* and *TPP4* were cloned into pK7FWG2 and pK7WGF2 (ref. 36), respectively, and transformed into the *Agrobacterium tumefaciens* strain GV3101. Tobacco infiltrations were performed as described³⁷ with *Agrobacterium* at an OD_{600} of 0.05, and a strain expressing p19 (ref. 38) was co-infiltrated with all constructs. After 2 d, leaves were stained with 1 μ g ml⁻¹ DAPI (4',6-diamidino-2-phenylindole; Thermo Fisher Scientific), and green fluorescent protein (GFP) and DAPI fluorescences were imaged using a Zeiss LSM 710 confocal microscope. For western blot, total protein was prepared from tobacco leaves³⁷ and GFP fusion proteins were detected using horseradish peroxidase-conjugated mouse anti-GFP antibody (Miltenyi Biotec).

Immunolocalization. Antibodies against *RA3* were generated in rabbits using a peptide consisting of the first 80 residues of *RA3* fused to a His-tag (produced in *E. coli*) as the antigen and purified using the same peptide³⁹. Immunolocalization was performed as described³⁹ on 2–3-mm ears harvested from 6-week-old *ra3* and B73 plants. *RA3* was detected using the purified primary antibody and Cy3-coupled goat anti-rabbit IgG secondary antibody (ImmunoResearch Laboratories). Nuclei were counterstained with 3C5 mouse monoclonal antibody^{40,41} and Alexa Fluor 568-coupled anti-mouse secondary antibody (Invitrogen). Imaging was performed on a Zeiss LSM 710 confocal microscope.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available from the corresponding author on request.

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Author contributions

H.C. performed all experimental procedures except for those listed below, prepared figures and co-wrote the manuscript. A.G. and S.L.V. isolated and fine-mapped *tpp4-1*. A.L.E. analysed the *tpp4-1* whole-genome sequencing data. X.X. mapped *tpp4-3* and *tpp4-4*. N.S.-N. performed immunolocalizations, under the supervision of H.S. R.F. performed metabolite measurements, under the supervision of J.E.L., who also co-wrote the manuscript. G.A.B. performed modelling of the TPP4 structure, supervised by R.G.B. D.J. supervised the research, assisted with mutant screening and co-wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Software and code

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Data collection

No software was used.

Data analysis

All statistical analyses were performed in R (v. 3.1+) using standard functions. Other software that was used: bwa-mem version 0.7.9; GATK UnifiedGenotyper version 3.0; Trimmomatic version 0.36; Hisat2 version 2.1.0, HTSeq version 0.6.0, edgeR version 3.22.0, Gramene release 56 (<http://www.gramene.org>), the Phyre2 web server (<http://www.sbg.bio.ic.ac.uk/~phyre2/>, accessed in February 2018).

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Sample size	No sample-size calculations were performed. Sample sizes typically used in the literature for these types of experiments were also used here.
Data exclusions	No data was excluded.
Replication	For all but one experiment at least two independent biological replicates were performed, in all cases showing similar results, and all performed replicates were included. The experiment shown in Supplementary Fig. 5 was not replicated as it provided a qualitative result that confirms generally accepted knowledge in the field, namely that TPPs have very high specificity for T6P as a substrate in plants and fungi (see e.g. Satoh-Nagasawa et al., 2006; Shima et al., 2007; Miao et al., 2016).
Randomization	For all experiments involving plant phenotyping or tissue collection from plants, segregating populations were used, and individuals were genotyped after planting in their final location. This ensured that all genotypes were randomized within each experiment.
Blinding	For all experiments involving plant phenotyping, genotypes were randomized (see randomization). Genotyping was then performed independently of any phenotyping, and the genotype was not known during phenotyping. Genotypes and phenotypes were only matched for analysis after all genotyping and phenotyping data had been collected. For other experiments blinding was not possible due to the nature of the experiments.

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involved in the study
☐	☒ Unique biological materials
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☒	☐ Human research participants

Methods

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☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Unique biological materials

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Obtaining unique materials All unique materials are available upon request from the corresponding author.

Antibodies

Antibodies used	Sigma mouse anti-HA (HA-clone 7, H9658, lot number 036M4796V, dilution 1:1,000) Miltenyi Biotec HRP-conjugated mouse anti-GFP (clone GG4-2C2.12.10, product number 120-002-105, lot number 5180102817, dilution 1:5,000) rabbit anti-RA3 (dilution 1:100) Cy-3 coupled goat anti-rabbit IgG (ImmunoResearch Laboratories, product number 111-165-003, lot numbers 67765 and 65175, dilution 1:200) 3C5 mouse monoclonal IgM (described in Turner et al., 1985 and Turner and Franchi, 1987) AlexaFluor 568-coupled goat anti-mouse IgM (Invitrogen, product number A21043, lot number 53641A, dilution 1:200)
Validation	anti-HA: This antibody has previously been used to detect HA-tagged proteins in yeast (see e.g. Wu et al., PLoS Genetics 2018), and has been cited many times. Moreover, we only detected a single band at the expected size, which was absent from the

empty vector control (the same yeast strain transformed with empty vector), confirming that the antibody specifically detected the tagged protein.

anti-GFP: This commercial antibody is commonly used to detect GFP. Moreover, the band at the expected size was absent from mock-infiltrated tobacco leaves, confirming that the antibody specifically detected the tagged protein.

anti-RA3: We validated the antibody using Western blot on protein extracted from young developing ears (the same tissue type used for immunolocalisation) of wild type B73 and the ra3 mutant, and a single band was seen at the expected size that was specific for the wild type and was absent in the ra3 mutant, confirming specificity to RA3. Similarly, no background was obtained in immunolocalisation experiments using ears from ra3 mutant plant (see Fig. 3).

Cy3-coupled goat anti-rabbit IgG and AlexaFluor 568-coupled goat anti-mouse IgM: these are well-characterised commercial antibodies, with many citations. Moreover, the absence of signal in immunolocalisation using ears of ra3 mutant plants (Fig. 3) confirms the specificity of the Cy3-coupled anti-rabbit secondary antibody.

3C5 mouse monoclonal IgM: this monoclonal antibody against SR-type splicing factors has been used in many studies and is widely cited, and was used in plants before to detect splicing factors (Fang et al., 2004).

Animals and other organisms

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Laboratory animals

Anti-RA3 antibody was generated in rabbits. Rabbit immunization and plasma collection was performed by Covance Inc. (Princeton, NJ, USA) in compliance with all relevant regulations, using two rabbits.

Wild animals

This study did not involve wild animals.

Field-collected samples

This study did not involve samples collected from the field.